

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical
Practical – V

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Date of Assignment: 31/07/26	Date/Time of Submission:31/07/26

Aim:- 2. Describing the Distribution of a Variable:

- a. Obtain a dataset and calculate descriptive statistics (mean, median, mode,variance, etc.) for a specific variable of interest.**

Code:-

```
import pandas as pd
df = pd.read_csv("Walmart_Sales.csv")
print(df.head())
df['Weekly_Sales']
print("Mean :", sales.mean())
print("Median :", sales.median())
print("Mode :")
print(sales.mode())
print("Minimum :", sales.min())
print("Maximum :", sales.max())
print("Range :", sales.max() - sales.min())
print("Variance :", sales.var())
print("Standard Deviation :", sales.std())
print("Count :", sales.count())
print("Sum :", sales.sum())
print("\n ishita patel")
print("\nDescriptive Statistics")
print(sales.describe())
```

Output:-

```
Store    Date    Weekly_Sales  Holiday_Flag  Temperature  Fuel_Price  \
0      1  05-02-2010    1643690.90           0         42.31      2.572
1      1  12-02-2010    1641957.44           1         38.51      2.548
2      1  19-02-2010    1611968.17           0         39.93      2.514
3      1  26-02-2010    1409727.59           0         46.63      2.561
4      1  05-03-2010    1554806.68           0         46.50      2.625

CPI  Unemployment
0  211.096358      8.106
1  211.242170      8.106
2  211.289143      8.106
3  211.319643      8.106
4  211.350143      8.106
Mean : 2749.509592666667
Median : 1902.42
Mode :
0      0.0
Name: Sales Revenue (USD), dtype: float64
Minimum : 0.0
Maximum : 27165.88
Range : 27165.88
Variance : 6597907.790099749
Standard Deviation : 2568.6392876579125
Count : 30000
Sum : 82485287.78
...
50%    1902.420000
75%    3863.920000
max     27165.880000
Name: Sales Revenue (USD), dtype: float64
```

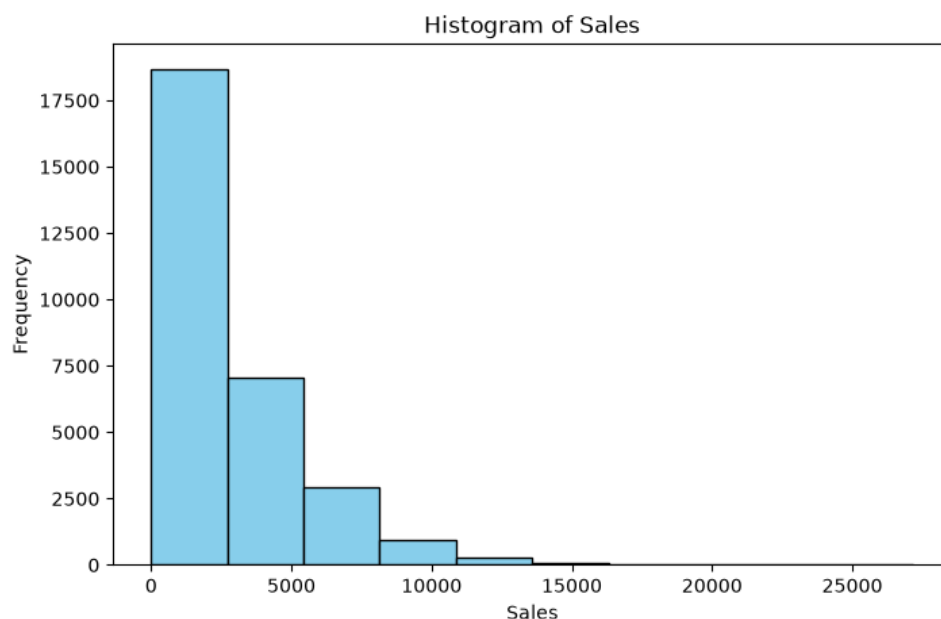
b. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

Histogram

Code:-

```
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(8,5))
plt.hist(sales, bins=10, color="skyblue", edgecolor="black")
plt.title("Histogram of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```

Output:-



Box plot

Code:-

```
plt.figure(figsize=(8,3))
sns.boxplot(x=sales, color="orange")
plt.title("Box Plot of Sales")
plt.xlabel("Sales")
plt.show()
```

Output:-

